Experiment 8: Stochastic Part-of-Speech Tagging in Python

1. Aim

To write a Python program to implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model.

2. Procedure

1. Define a tagged training corpus containing word and POS tag pairs.
2. Count the frequency of each tag per word using `defaultdict` and `Counter`.
3. Determine the most probable POS tag for each known word based on maximum likelihood.
4. Assign the default tag `'NN'` for unseen words.
5. Display the stochastically tagged output for a test sentence.

3. Program

```
from collections import defaultdict, Counter

corpus = [("The", "DT"), ("dog", "NN"), ("runs", "VBZ"), ("runs", "VBZ"), ("runs", "NNS")]
counts = defaultdict(Counter)

for word, tag in corpus:
    counts[word][tag] += 1

words = ["The", "dog", "runs", "fast"]
tagged = [(w, counts[w].most_common(1)[0][0] if w in counts else "NN") for w in words]
print("Stochastic POS Tags:", tagged)
```

4. Output

```
Stochastic POS Tags: [('The', 'DT'), ('dog', 'NN'), ('runs', 'VBZ'),
('fast', 'NN')]
```

5. Result

Thus, the Python program was successfully executed to implement a simple stochastic part-of-speech tagging algorithm using a probabilistic model.